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A detailed rainfall climatology for Malawi, Southern Africa

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ABSTRACT: This article presents a detailed analysis of the mean rainfall climatology of Malawi and an overview of interannual variability. Monthly and seasonal means, the seasonal cycle and intraseasonal variability are examined. This includes an analysis of the onset, end and duration of the rainy season. The study reveals two surprising features of the rainfall regime. One is a region with a strong rainfall maximum in the austral autumn along the western shore of Lake Malawi. The second is a brief period of reduced rainfall in mid-February that appears to signal a shift in the prevailing rainfall and circulation regime. The prevailing atmospheric circulation is markedly different before and after the break, as are the characteristics of the rainy season. Overall, the results indicate that the early (pre-break) rainy season (December–February) is dominated by tropical influences, the late (post-break) rainy season (March–April) is dominated by extra-tropical influences and interannual variability in these two periods is uncorrelated. The results also demonstrate two aspects of interannual variability: the dominant contribution of the MA season and the frequent opposition between northern and southern Malawi.

KEY WORDS Southern Africa; rainfall; intraseasonal variability; interannual variability

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1. Introduction

A major focus in current climate studies is how climate will change in response to global warming. Model simulations for the mid-21st century suggest that the impact will be greatest in Africa (IPCC, 2007). The few available projections suggest that Southern Africa, in general, will experience drier and more extreme conditions and an increased frequency of drought (Wang, 2005; Moise and Hudson, 2008; Shefield and Wood, 2008). More favourable conditions are projected for the equatorial latitudes (Shongwe *et al.*, 2011). For Malawi, lying in the transition zone between these two regions, the projections are ambiguous, with some models predicting higher rainfall and others (Shongwe *et al.*, 2009) suggesting opposite changes in the north and south of the country.

Other factors further pose difficulties in projecting the impact of climate change on Malawi. One is the potential role of regional factors such as Lake Malawi and the high terrain surrounding it (Figure 1). Just to the north, this same combination of geographical factors has a strong influence on the precipitation regime around Lakes Victoria and Tanganyika (Nicholson and Yin, 2002). Similarly, the highlands surrounding the Zaire Basin are a dominant control on the precipitation and convective regimes over the basin (Jackson *et al.*, 2009).

The effect of these regional features on Malawi's rainfall regime is largely unknown because only a few studies (Ngongondo *et al.*, 2011a, 2011b) have taken a detailed look at the country's rainfall climatology. In this article, we do so indirectly by examining both the temporal and spatial variability of rainfall in Malawi and the atmospheric circulation features associated with the rainy season. In doing this, we also examine to what extent rainfall variability is coherent throughout the country and to what extent the characteristics of the rainfall regime are uniform throughout the season.

The results reveal two surprising features of the rainfall regime. One is a region with a strong rainfall maximum in the austral autumn along the western shore of Lake Malawi. The second is a brief period of reduced rainfall in mid-February that appears to signal a shift in the prevailing rainfall and circulation regime. These features are examined in detail.

The article commences with a basic description of the mean rainfall climatology and a delineation of four homogeneous rainfall regions within Malawi (Section 3) and follows with a description of the seasonal cycle (Section 4). Then interannual variability is examined, using century-long time series for these regions and a two-century long time series for the country as a whole (Section 5). The question of a mid-season regime change is evaluated by examining the atmospheric circulation before and after the mid-February break (Section 6). Finally, the characteristics of intraseasonal and interannual variability are contrasted for December–February (DJF) and March–April (MA), i.e. the early and late rainy season (Section 7).

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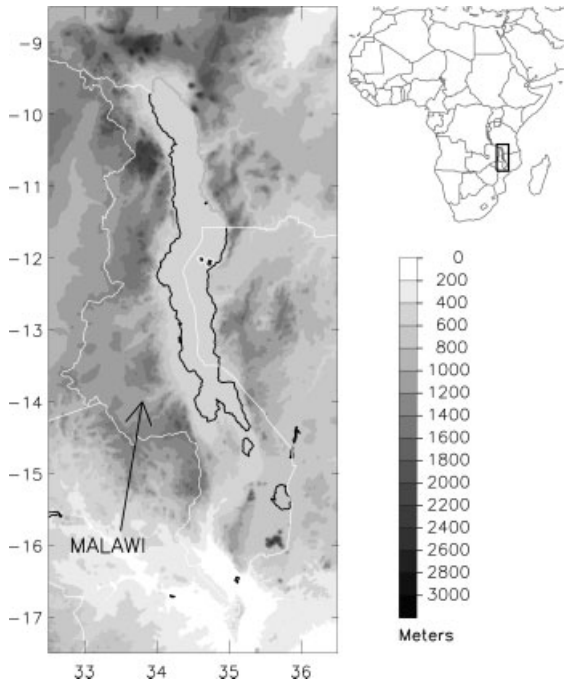

Figure 1. Topographical relief over Malawi and surrounding regions (from NOAA: www.ngdc.noaa.gov/mgg/topo/globe.html). Lake Malawi is outlined in black.

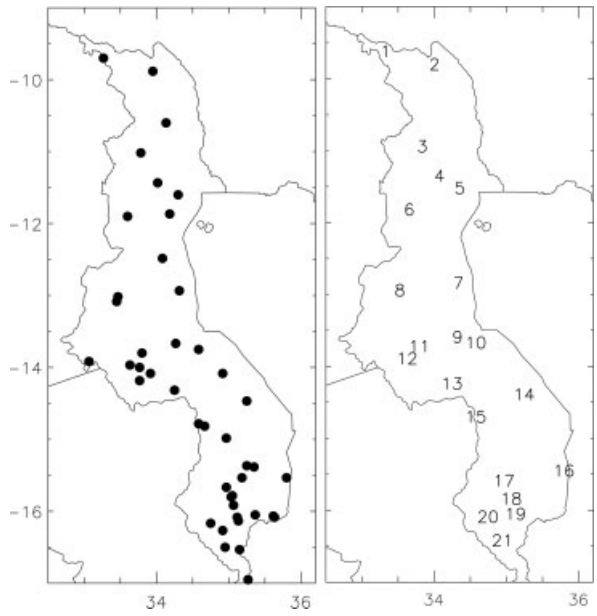

Figure 2. Rainfall stations used in this study. Left: long term, monthly data available. Right: daily data available.

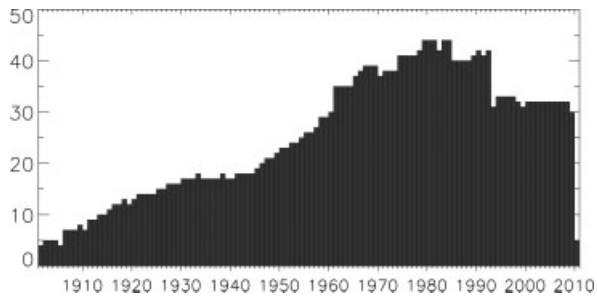

Figure 3. Number of stations per year in the first author's archive.

2. Data and methodology

The precipitation data utilized in this analysis were generously provided by the Meteorological Service of Malawi. Quality control of the data was carried out by that agency. Malawi has an excellent network of rainfall stations, some of which extend back to the 19th century. This study uses a select set of these, some 75 stations well distributed over the country. For each station, mean dekadal (10 day) totals are available for the rainy season months. For a small group of stations, monthly totals are available for individual years (Figure 2). For this group, the record length varies, so that the number of available stations ranges from 4 to 5 in the 1900s to 40 to 45 in the 1970s and 1980s (Figure 3). Monthly, seasonal and annual means are calculated at each station over the length of record. Although the record length varies, this gives a better picture of mean conditions than does any 30-year period because of the relatively large interannual variability in this region. For a subset of 35 stations, daily data are also available. From these, we have selected for daily rainfall analysis 21 with nearly complete records for the period 1962–2009. These stations plus their coordinates are given in Table 1 and their locations are shown in Figure 2.

Preliminary analyses suggested that the rainfall regime might not be homogeneous throughout Malawi. To determine if that is the case, and if necessary, to identify homogeneous regions, each station was correlated with every other. On the basis of a subjective evaluation of the results, it was concluded that four regions should be delineated, combining the stations as shown in Figure 4. The number of stations in each region varies from 5 to 19.

An F-test demonstrated that each of these regions is relatively homogeneous with respect to interannual variability. The test examines the ratio of spatial variability V_{area} to temporal variability V_{time} (Kraus, 1977; Nicholson, 1986a). F ratios exceeding critical limits, based on the degrees of freedom in the analysis, indicate that a given region can be treated as a single entity for analyses of interannual variability. The F ratio for each region is shown in Table 2. These ratios well exceed those corresponding to the 1% significance level, indicating coherent variability within each of these regions.

The inter-regional correlations confirm the necessity of utilizing several regions within Malawi (Table 3). The correlations range roughly from 0.4 to 0.65. Thus, each regional series captures at most 15–42% of the interannual variability of the other regions. For the DJF season, the inter-regional correlations are generally lower, ranging from 0.27 to 0.56. For the MA season they are somewhat higher, ranging from 0.42 to 0.74.

Annual and seasonal time series are calculated for each of the four regions, following the methodology of Nicholson (1986a). Rainfall is expressed as a standardized anomaly (departure from the mean divided by the

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Table 1. The 21 rainfall stations used in the daily analysis, including station coordinates and elevation in metres, region to which assigned and station number.

Name	Latitude	Longitude	Elevation	Region	Number
Karonga	−9.88	33.95	529	1	1
Mzuzu	−11.43	34.02	1254	1	2
Nkhata Bay	−11.60	34.30	500	1	3
Nkhota Kota	−12.92	34.28	500	1	4
Chitipa	−9.70	33.27	1285	2	5
Bolero	−11.02	33.78	1100	2	6
Mzimba	−11.90	33.60	1349	2	7
Kasungu	−13.02	33.47	1058	2	8
Chitala	−13.67	34.27	606	3	9
Salima	−13.75	34.58	512	3	10
Lilongwe	−13.80	33.80	1229	3	11
Chitedze	−13.97	33.63	1149	3	12
Dedza	−14.32	34.25	1632	3	13
Mangochi	−14.47	35.25	482	3	14
Ntcheu-Nkhande	−14.78	34.58	1158	3	15
Makhanga	−15.53	35.80	52	4	16
Chileka	−15.67	34.97	767	4	17
Bvumbwe	−15.92	35.07	1146	4	18
Thyolo	−16.13	35.13	820	4	19
Chikwawa	−16.17	34.75	95	4	20
Ngabu	−16.50	34.95	102	4	21

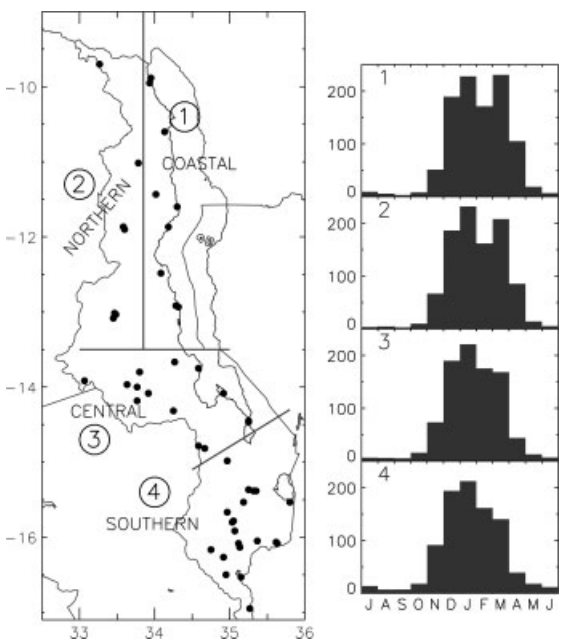

Figure 4. Left: four homogeneous rainfall regions of Malawi and the stations within them. Right: The typical seasonal cycle of rainfall (mm per month) in each.

standard deviation). A regional anomaly for the year or season is calculated as the average over all available stations in the region.

3. Mean climatology

Figure 5 shows mean annual and seasonal rainfall over Malawi. Throughout the country, mean annual rainfall is in the order of 800 mm to over 1600 mm. For most of the country, the austral summer is the rainiest season, with seasonal means on the order of 600–800 mm. Rainfall is relatively evenly distributed during these months. The MAM season is notably drier, but the distribution shows two strong, local maxima in the extreme north and in central Malawi. Seasonal means reach 600–700 mm in these maxima. The austral winter (JJA) is exceedingly dry over most of Malawi, with seasonal rainfall on the order of 10–25 mm. The spatial distribution is similar to that in MAM, but the localized maxima only reach 75 mm for the 3 months. The September to November (SON) season is also quite dry, with seasonal means on the order of 50–75 mm.

Figure 5 appears to show a rainfall maximum over the lake. It should be noted that the contours over the lake are extrapolated from land-based gauge data. However, an analysis of satellite data (Nicholson and Yin, 2002) confirms the maximum over the lake and along its western shore. This maximum appears to be related to topographic effects, as opposed to the effects of the lake itself, because the annual means for over-lake and over-land rainfall are similar. This contrasts strongly with Lakes Victoria and Tanganyika, which respectively enhance rainfall 35 and 11% (Nicholson and Yin, 2002).

Table 2. Determination of regional homogeneity by F ratio. Included are number of stations, variance in time and space, degrees of freedom in time and space, the F ratio and the F ratio corresponding to the 1% significance level.

Region	1	2	3	4
No. of stations	7	5	13	19
V_{time}	2.1	1.9	3.7	5.5
V_{area}	0.65	0.64	0.52	0.51
df_{time}	336	208	604	986
df_{area}	105	83	106	109
F ratio	3.3	3.0	7.2	10.9
F ratio (1%)	1.5	1.6	1.4	1.4

Table 3. Inter-regional correlations for the DJF and MA seasons and for annual totals.

Region	1 versus 2	1 versus 3	1 versus 4	2 versus 3	2 versus 4	3 versus 4
DJF	0.41	0.35	0.27	0.56	0.34	0.53
MA	0.42	0.48	0.50	0.62	0.45	0.74
Annual	0.57	0.54	0.40	0.65	0.46	0.63

4. The seasonal cycle

Figures 4 and 5 demonstrate that Malawi rainfall is concentrated in the months of December–March. In three of the four regions, maximum rainfall occurs in January. In the region along the western lakeshore, the maximum occurs in March (Figure 4). At stations in the northern region, to the west of this, a strong peak is evident in March, but it is less pronounced than the January peak.

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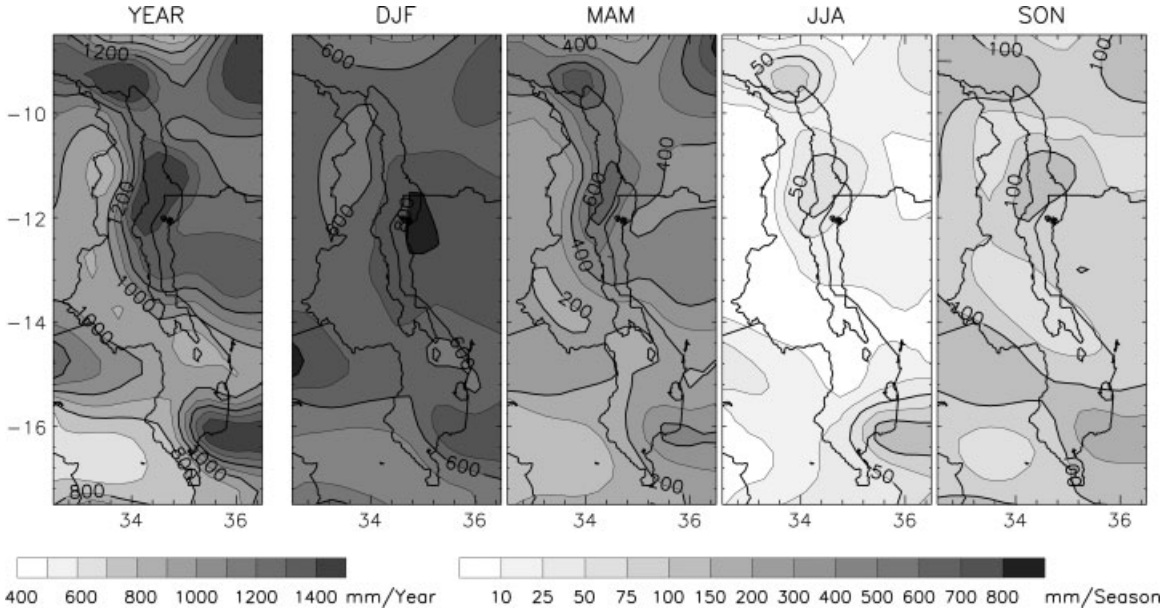

Figure 5. Mean annual and seasonal rainfall in mm (based on the period 1962–2009).

The seasonality of rainfall is very pronounced. Over most of the country 60–70% falls in the wettest quarter (Figure 6(a)). This increases to 70–80% in central Malawi, but falls to 50–60% just to the west of Lake Malawi (Figure 6(a)). At stations near the lake 40–50% of the annual rainfall occurs in March and April (Figure 6(b)); March alone contributes roughly 20–25% of the annual rainfall (Figure 6(c)).

Figure 7 shows the seasonal cycle at each of 75 stations for which monthly data are available. Calculations are based on the period 1971–2000. Although it is hard to distinguish individual stations from the graph, this plot serves to demonstrate that the monthly distribution of rainfall is relatively uniform throughout Malawi except during the months of March and April. This suggests a strong influence of regional factors during these months. A handful of stations exhibit strong peaks in MA, with some stations receiving most of their rainfall during those 2 months. The standard deviation of rainfall across the array of stations quantifies the contrast between March and April and other rainy season months. In the months of DJF, it is in the order of 30–40 mm, or roughly 20% of the monthly mean. In March it is 70 mm, or 39% of the mean. The months of November and April have comparable rainfall, but the standard deviation of station means is 25 (36%) in November but 66 mm in April (89%).

To obtain a more detailed view of the start and end of the rainy season, dekadal (10 day) averages were examined. The results for 18 stations are shown in Figure 8. These indicate that the rainy season generally commences in late October or early November and runs through early April. Following Tadross *et al.* (2009), the onset and end of the rainy season were quantified using the following definitions. The season has commenced once 25 mm of rainfall has accumulated within 10 d, without 10 consecutive dry days (<2 mm) occurring afterward. The caveat removes from the calculation so-called 'false starts' of the season. The end is defined as three consecutive dekads (after February 1) of <20 mm each.

These statistics were calculated for the 21 stations (Figure 2) with more or less continuous daily records for the period 1964–1965 to 2008–2009. Table 4 gives the average onset and end dates and season duration for each station. The onset is relatively uniform over Malawi, ranging from mid-November to early December. Except for stations in region 1 (near the lake), the end is generally in mid-March to early April. In region 1, the season persists until mid- to late April or early May. The typical duration is in the order of 105–125 d. Region 1, where the season is as long as 167 d, is again an exception, as are two stations in the far south of region 4.

Figure 8 also shows that at most stations a small decrease in rainfall occurs in mid-season, on average during the sixth dekad of the year (days 51–60). A similar situation occurs in Botswana, where a pronounced break occurs in late December/early January throughout most of the country. Bhalotra (1984) suggests that the break represents the shift between tropical and extra-tropical control on Botswana rainfall. We hypothesize that the late February break in the Malawi rainy season likewise represents a regime shift. This hypothesis is examined in Sections 6 and 7. It is noteworthy that at stations in northern Botswana, a second break occurs at the same time as that in Malawi.

5. Interannual variability

5.1. Long-term variations in Malawi rainfall

Figure 9 shows annual rainfall in the four regions and for the country as a whole. Following Nicholson (1986a) and others, rainfall is expressed as a regionally averaged

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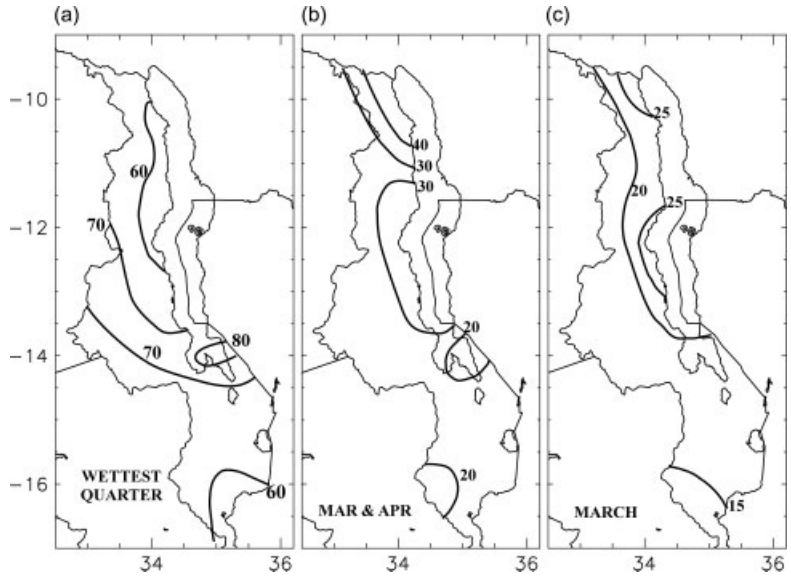

Figure 6. (a) Percent concentration of rainfall in the wettest quarter of the year. (b) Percent of annual rainfall occurring in March. (c) Percent of annual rainfall occurring in March plus April.

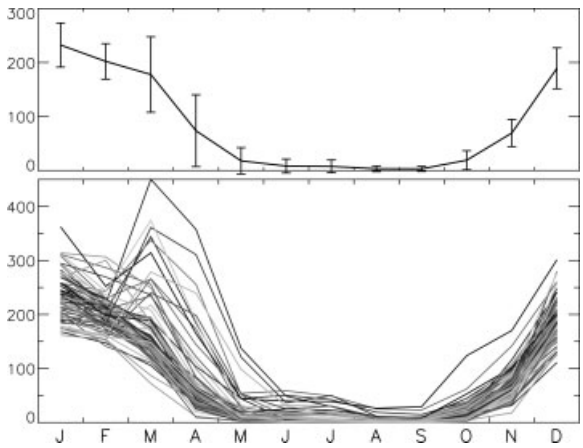

Figure 7. Bottom: The seasonal cycle at 75 stations in Malawi. Top: The monthly means (mm) averaged over the array of stations and the standard deviation of the station means (mm, vertical bars).

standard departure. No long-term trends over the course of the analysis period are apparent. However, in the two northern regions (1 and 2) rainfall has generally been below normal during the last two decades. During these decades, several intense drought years occurred in the southern regions (3 and 4). For the country as a whole, rainfall was frequently below average during the first few decades of the 20th century. This tendency is most strongly apparent in region 1 (the northwestern lake shore).

Figure 10 shows Malawi rainfall variability from the early 19th century to the present. This is based on a newly created semi-quantitative data set that combines documentary and gauge records (Nicholson *et al.*, 2012a). Rainfall is expressed in a seven-class system (−3 to +3). From 1901 onwards, the time series is based solely on gauge records converted to this system.

The contrast with the 20th century is striking. Intense drought prevailed throughout most of the period 1801 through the early 1860s. The droughts of the 1820s and 1830s affected nearly all of Africa (Nicholson *et al.*, 2012a, 2012b) and left clear evidence in major lakes throughout Africa (Nicholson, 1998a, 1998b, 1999, Verschuren *et al.*, 2000, Shanahan *et al.*, 2009). Lake Malawi was some 10 m below its modern stand (Nicholson, 1998a). The early 19th century stands out not only for the magnitude and frequency of drought, but also its persistence. In the modern record, more than two consecutive years of below average rainfall occurred only infrequently. The longest period with below average rainfall was 6 years (Figure 9, bottom), compared to 21 in the early 19th century (Figure 10).

5.2. Spatial modes of variability

Figure 9 suggests that Malawi is not strongly homogeneous with respect to interannual variability. Particularly striking is the contrast between region 1 and the remaining regions. However, for annual rainfall the correlations among the four regions (Table 3) are all positive and significant, ranging from 0.40 to 0.65. The strongest similarity is between regions 2 and 3 and 3 and 4, the central and southern regions. The inter-regional correlations are somewhat higher for the MA season, ranging from 0.42 to 0.74. Notably, they are considerably lower for the DJF season, ranging from 0.27 to 0.56.

A principal component (PC) analysis, using the 21 long-term stations and the time period 1961–1997, confirms a high degree of coherence within Malawi. The first two PCs (Table 5) explain a total of 54% of the variance. The first, which explains 43% of the variance, has anomalies of the same sign throughout the country (Figure 11). Loadings vary from roughly 0.4 to 0.8 at the individual stations. The second PC, which explains

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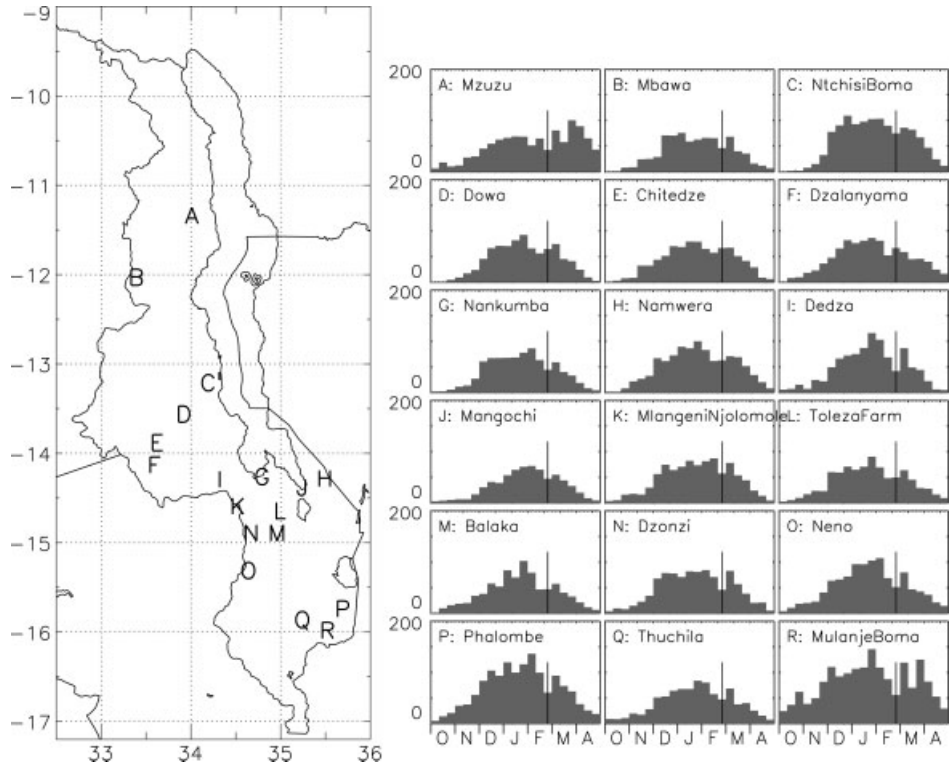

Figure 8. Mean dekadal (10 day) rainfall in mm at selected stations in Malawi. The vertical lines indicate the timing of the February break. The location of stations is shown on the left.

Table 4. Mean onset, end and duration of the rainy season for individual stations in regions 1–4. See Table 1 for identification of stations.

Region	1	1	1	1	2	2	2	2
Station	1	2	3	4	5	6	7	8
Onset	12/4	11/19	11/23	12/1	11/27	12/6	11/27	12/1
End	4/20	4/28	5/8	4/14	4/4	3/19	3/30	3/20
Duration	138	161	167	134	129	104	124	108

Region	3	3	3	3	3	3	3	4	4	4	4	4	4
Station	9	10	11	12	13	14	15	16	17	18	19	20	21
Onset	12/4	12/3	11/28	11/27	11/27	12/4	11/23	11/27	11/15	11/14	11/13	11/29	11/25
End	3/24	4/1	3/21	3/23	3/27	3/19	3/27	3/17	3/17	4/4	4/5	3/13	3/19
Duration	111	120	114	117	122	106	126	111	123	142	144	105	114

11% of the variance, shows anomalies of the opposite sign in northern and southern portions of the country.

A PC analysis based on regionally averaged rainfall data for the sector 5°S and 25°S and the years 1920–1994 suggests that this opposition is the most common pattern of regional-scale climatic variability. In this analysis, the homogeneous regions delineated in Nicholson (1986a) are utilized. The results show that at this scale, the first three PCs explain 56% of the variance (Table 5). The first two provide evidence of a discontinuity through central Malawi (Figure 12). PC1 shows predominantly positive anomalies, but they become very weak or negative around 10°S–15°S. The opposition is more striking in PC2, with strong loadings of the opposite sign to the north and south of ca 15°S.

It is well known that the patterns shown can be artefacts of the orthogonality constraint of PC analysis. However, the same patterns were apparent in earlier analyses based on a method that is free of this constraint (Nicholson, 1986a, 1986b). The opposition between equatorial and subtropical latitudes is strongly apparent. During the period 1901–1973, the discontinuity separating regions of positive and negative anomalies extended through Malawi in roughly 1 of 3 years.

6. Atmospheric circulation in the early and late rainy season

The decrease in rainfall noted in the sixth dekad at most Malawi stations suggests a transition between

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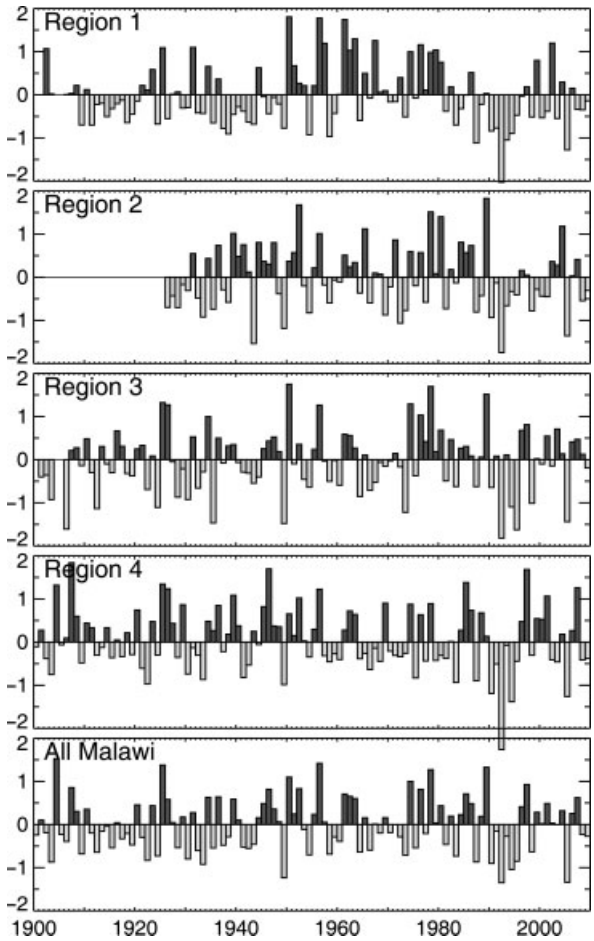

Figure 9. Annual rainfall 1901–2003 for four regions of Malawi and for the country as a whole. Rainfall is expressed as a standardized departure, following Nicholson (1986a).

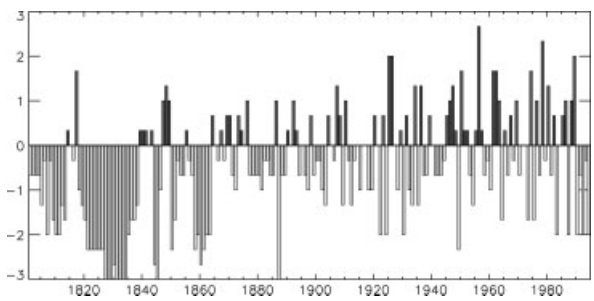

Figure 10. Annual rainfall for Malawi since 1801, expressed as a rainfall index ranging from −3 to +3 (Nicholson 2001; Nicholson *et al.*, 2012a, 2012b).

two different atmospheric regimes. Notably, the months following the sixth dekad March and April, are part of the main rainy season in the equatorial latitudes north of Malawi. Those preceding it, DJF, are the main rainy season in the latitudes south of Malawi. Such a situation can have important consequences for understanding and predicting climate variability and change in the region.

To determine whether the brief February break relates to a major shift in circulation features, the wind regimes during the four dekad prior to and after the sixth dekad

Table 5. Percent variance explained by the first five principal components for the Malawi and subtropical geographical regions.

PC number	1	2	3	4	5
Malawi	43	11	7	6	5
Subtropics	28	15	13	7	6

are examined. These are henceforth referred to as the early and late rainy season, respectively. The circulation at 925, 850, 700, 600 and 200 mb is examined and vertical cross-sections of zonal winds as a function of latitude and longitude are constructed. Throughout the troposphere striking contrasts are apparent between the early and late rainy season.

At 925 mb, the winds are weak over Malawi in the early rainy season; strong northeasterlies prevail to the northeast and southeasterlies prevail in the southwest (Figure 13). Wind convergence is generally weak. After the break, the winds are generally stronger and predominantly easterly or southeasterly and convergence prevails around 12°S. Similar contrasts are apparent at 850 mb (not shown). In the mid-troposphere at both 600 and 700 mb, there is a pronounced reversal in the wind direction around 10°S–15°S, with westerlies in the early rainy season and easterlies in the late rainy season. At 200 mb, the tropical easterlies are stronger and further south before the break than afterward, when circulation features shift towards the Northern Hemisphere and the subtopical westerlies are apparent from 17°S–20°S. This regime allows for strong mid-latitude influence on the rainfall regime.

Mean zonal winds before and after the sixth dekad were examined along east–west transects at 10°, 12.5° and 15°S and along north–south transects at 32.5°E and 35°E. The three sets of east–west transects show similar results, so that only the transect at 12.5°S is shown here (Figure 14(a)). For the same reason, only the north–south

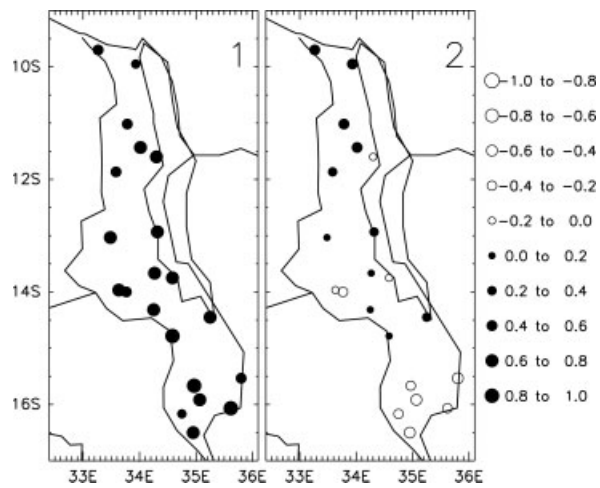

Figure 11. Spatial loadings of the first two principal components for Malawi rainfall, based on station data.

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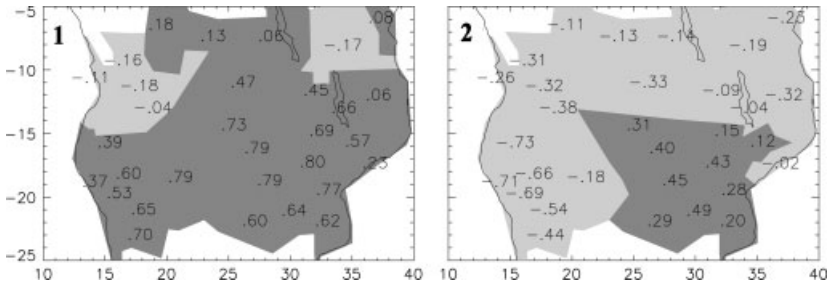

Figure 12. Spatial loadings of the first two principal components for equatorial rainfall based on spatially averaged data for the regions delineated in Nicholson (1986a).

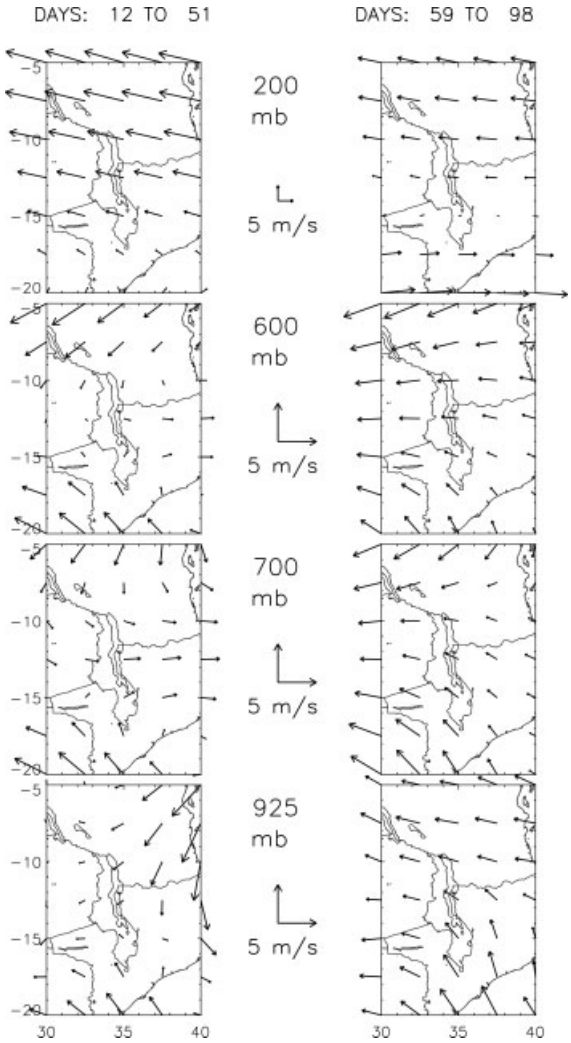

Figure 13. Mean vector winds in the early and late rainy season at 925, 700, 600 and 200 mb.

transect at 35°E is shown (Figure 14(b)). These highlight the patterns evident in the vector winds.

During the early rainy season, the Tropical Easterly Jet (TEJ) is dominant in the upper troposphere and a strong core of westerlies is apparent in the mid-troposphere. The maximum westerly wind is at 700 mb and 12°S–13°S. In the late rainy season, the TEJ is weaker and displaced equatorward. The subtropical westerlies have likewise shifted equatorward and are strongly evident poleward of 15°S. The subtropical westerlies are replaced by a weak easterly jet centred at 850 mb and 8°S. This shift from easterlies to westerlies in the upper troposphere suggests that in the early rainy season a tropical regime prevails, shifting to a dominant extra-tropical in the late rainy season.

The development of easterly flow in the mid-troposphere and intensification of the easterlies at 850 mb in the late rainy season are the logical reasons for the MA maximum along the western shore of Lake Malawi. Orographic uplift would result when the flow encounters the highlands paralleling the lake on both its eastern and western sides.

7. The early and late rainy season: interannual and interseasonal variability

Section 6 demonstrates strong contrasts in the atmospheric circulation regimes in the early and late rainy season and a likely shift from tropical to extra-tropical dominance. This suggests that the characteristics of interannual and intraseasonal variability probably also differ in the early and late season. Hence, these characteristics are examined separately for DJF and MA. The analysis includes interannual variability over the course of the 20th century and the number and magnitude of rain events.

7.1. Interannual variability of seasonal rainfall

Figures 15(a)–(d) show rainfall for the DJF and MA seasons and annual rainfall for the July–June period. Rainfall in both seasons exhibits year-to-year fluctuations that show a strong similarity to fluctuations of annual rainfall. However, linear correlation (Table 6) shows some contrasts between the regions. For region 1, the correlation between DJF and annual rainfall is 0.71 for the period 1903–2010, but the correlation between MA and annual rainfall is 0.81. Thus, the MA season contributes the lion's share of the interannual variability, or roughly 66%, but it contributes only about 40% of the annual mean. For the remaining three regions, the correlation between DJF and annual rainfall is notably higher than the correlation between MA and annual rainfall. The latter range from 0.62 to 0.66, suggesting that MA contributes roughly 40% of the interannual variability. In

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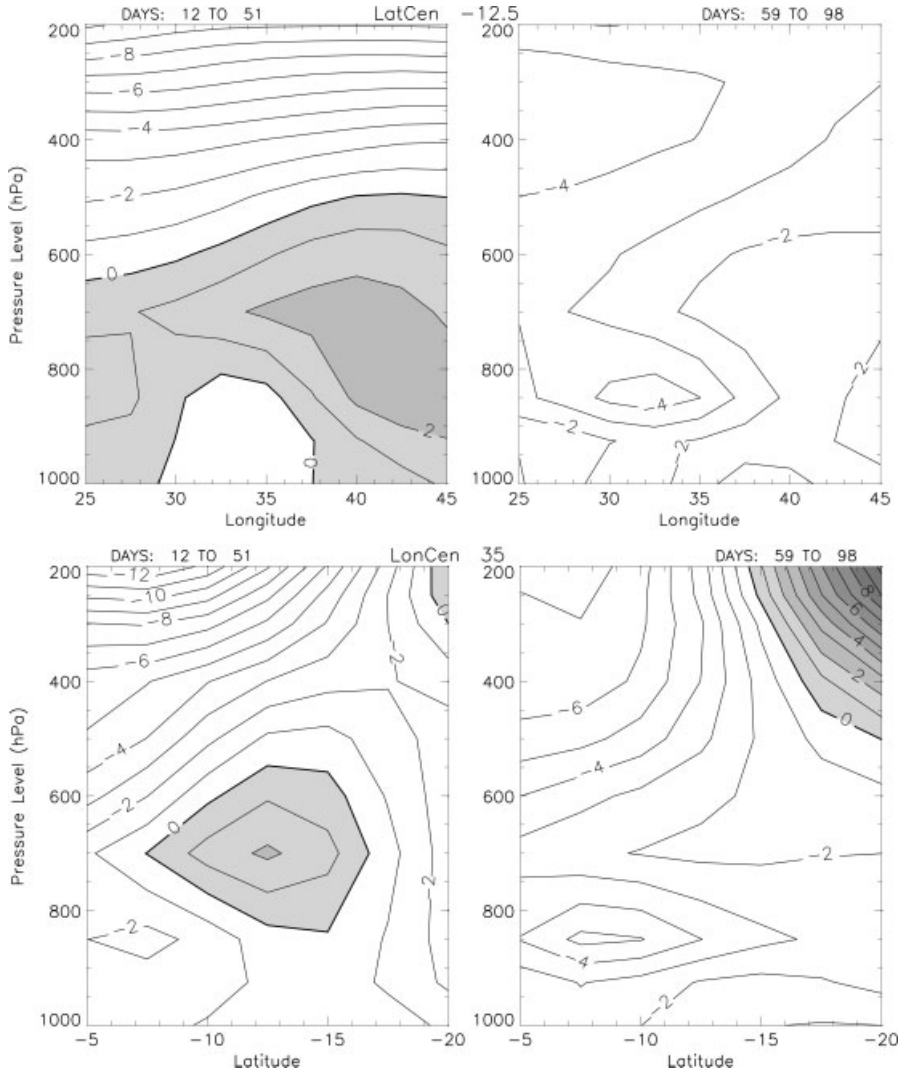

Figure 14. Mean zonal wind speed ($m\,s^{-1}$): latitudinal transect 35°E (bottom) and longitudinal transect at 12.5°S (top) for the early and late rainy seasons (days 12–51 and 59–98, respectively). Westerly winds are shaded.

comparison, these months contribute roughly 20–25% of annual rainfall at most stations in these regions. Hence in these regions as well the contribution of MA rainfall to interannual variability is considerably higher than the contribution to mean annual rainfall.

Notably, the interannual variability in the early and late seasons (DJF and MA, respectively) is relatively uncorrelated. The highest correlation is for region 1 and it is only 0.30; for region 2 it is only 0.12. These low correlations further support our hypothesis that two different regimes prevail in the early and late rainy season.

7.2. Frequency and intensity of rain events in the early and late rainy season

The early and late season also contrast in terms of the number and magnitude of rain events, as assessed from daily rainfall totals. Table 7 gives the mean number of rain events per month and the mean intensity of the event (amount of rainfall divided by the number of rain days) for individual stations in regions 1–4. These statistics are tabulated separately for DJF and MA.

The statistics demonstrate contrasts not only between the early and late rainy season but also confirm the contrasts between region 1 and the other three regions. In region 1, roughly the same number of events per month occurs in DJF and in MA. However, in MA, the mean intensity is greater. In the remaining regions, there are far fewer rain events per month in MA than in DJF and the mean intensity is lower.

8. Summary and conclusions

This study has documented three aspects of Malawi's rainfall regime that have not been previously described. Perhaps, the most important is the mid-season 'break' and the associated contrasts in the circulation and rainfall regimes prior and subsequent to it. This break, a brief reduction in mean dekadal rainfall, occurs during the sixth dekad (late February) and is evident at nearly all stations in Malawi. The break appears to mark the transition between the dominance of a tropical rainfall

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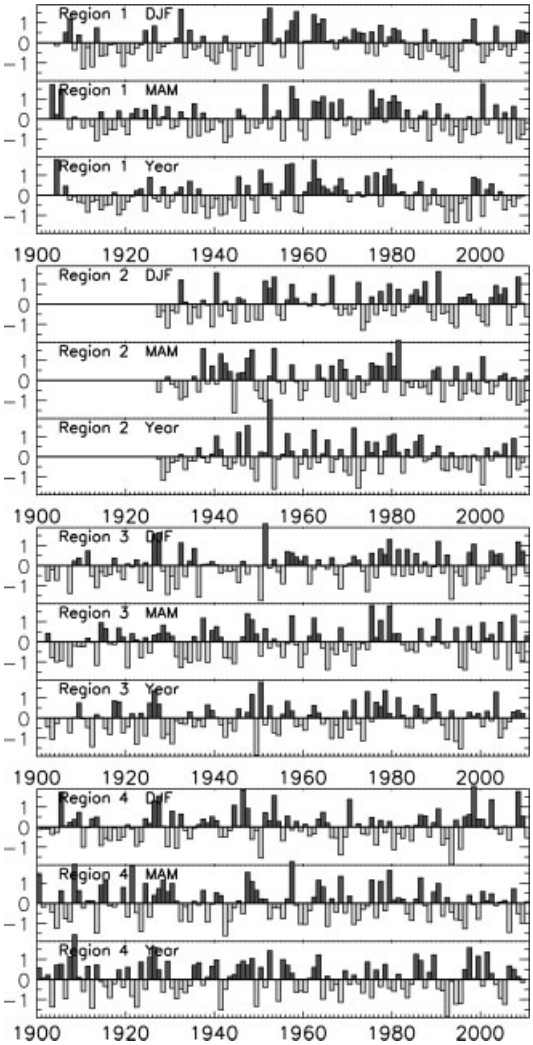

Figure 15. Time series of rainfall for the DJF and MA seasons and for the agricultural year (July–June). These are shown for each of the four regions and are expressed as standardized departures from the mean. The annual total is plotted for the year corresponding to the January–June period.

regime in the early rainy season and a mid-latitude regime in the late rainy season. Interannual variability is markedly different in the early and late rainy season.

The second feature is a region in which the majority of rainfall occurs during March and April. This region includes stations along the western shore of Lake Malawi, but probably extends over the lake itself in its central and northern portions. In this region, rainfall is also high in DJF. Thus, this is one of the wettest parts of the country and the area with the longest rainy season. A related feature is the prevalence of nocturnal rainfall over the region and in the highlands to the west of it (Nicholson and Yin, 2002), suggesting a strong orographic influence. That conclusion is supported by the change in atmospheric circulation that occurs between February and March. Strong easterlies develop in the mid-troposphere, forcing orographic ascent when they reach the highlands west of Lake Malawi.

Table 6. Correlation between the DJF and MA seasons and for both seasons with annual rainfall, for Regions 1–4.

Region	DJF versus MA	DJF versus Year	MA versus Year
1	0.30	0.71	0.82
2	0.12	0.79	0.62
3	0.24	0.86	0.64
4	0.16	0.76	0.66

A third feature we demonstrate is the importance of the MA season. For region 1 MA contributes the lion's share of the interannual variability. The correlation with annual rainfall is 0.81 during the period 1903–2010. Thus, these 2 months account roughly for 66% of the interannual variability, but they account for only about 40% of mean annual rainfall. For the remaining three regions, the correlations range from 0.62 to 0.66, suggesting that MA contributes roughly 40% of the interannual variability. In comparison, these months contribute roughly 20–25% of annual rainfall at most stations in these regions. Hence, the contribution of MA rainfall to interannual variability is considerably higher than the contribution to mean annual rainfall. Its importance is probably linked to El Niño, the impact of which is greatest during March through May in this area (Nicholson and Kim, 1997).

Our work has also demonstrated the spatial heterogeneity of Malawi's rainfall regime. Four distinct regions have been identified. Those in central and southern Malawi have a single rainfall maximum in the seasonal cycle and the two northern regions have a double rainfall maximum. Inter-regional correlations suggest that for the year as a whole any two regions have at most 42% common variance on interannual time scales. Generally, the common variance is about 15–25%. A PC analysis indicates that year-to-year fluctuations are frequently of the opposite sign in northern and southern portions of the country.

The inter-regional correlations indicate that interannual variability is more spatially coherent during the late rainy

Table 7. Mean number of rain events (i.e. rain days) per month and mean rainfall intensity (mean amount per rain event) in the DJF and MA seasons.

Region	1	1	1	1	2	2	2	2	3	3	3	3	3	3	3	4	4	4	4	4	4
Station	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
#DJF	17	20	18	17	19	15	18	16	15	17	18	18	19	15	16	11	15	18	17	10	12
#MA	18	20	19	15	13	9	12	8	8	10	10	10	11	9	8	7	9	14	14	6	8
DJF (mm)	11	10	12	16	11	10	11	12	14	17	11	11	12	12	15	14	12	13	14	15	14
MA (mm)	14	11	17	18	9	8	9	9	11	17	9	9	8	9	12	9	9	9	10	11	9

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season of MA than during the early rainy season of DJF. However, it is still relatively low during MA. The common variance shared by any two regions is less than 40%, except in the case of regions 3 and 4, which are relatively dry during March and April. On the other hand, the spatial variability of monthly rainfall is much greater in March and April than during other months.

A major implication of our results is that the factors governing interannual variability may be quite different for the early and late rainy seasons. Model projections of climate change in Malawi and seasonal forecasts for that country must separately evaluate these two periods. Also, the dominance of the late rainy season in the areas near and possibly over the lake suggest that global climate change might affect these region differently than the rest of the country. Such a situation may exist in other locations as well. Further studies of this type could provide the relevant insight.

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